(Please write your Enrollment Number) Enrollment No. 15101172022

End-Term Examination (Regular & Reappear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech IT/ CSE/ MAE/ ECE/ CSE-AI/ ECE-AI/ DMAM>, Semester:<3rd>
(November-December, 2023)

Subject Code: BCS 201	Subject: Data Structures
Time :3 Hours	Maximum Marks :60

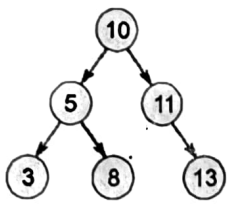
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) Assume that a binary search tree with 1000 unique data is implemented using array representation. It is also known that the constructed binary search tree forms a complete binary tree. If the array indexes start with 0 then find the index of the 3rd largest element of the binary search tree.

(b) Consider the given below C function mystery and the binary tree :

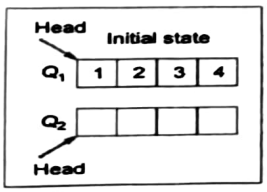
```
typedef struct node
{ int val; struct node *lchild, *rchild;
} node;
int mystery (node *p) {
 int retval;
 if (p==NULL) return 0;
 else {
 retval= p->val + mystery (p-> lchild) + mystery (p-> rchild
);
 printf ("%d", retval);
 return retval;
}}
```

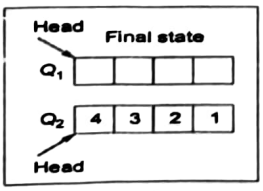


When mystery is called with a pointer to the root node of the given binary tree, what will it print?

(c) What is the worst-case time complexity of inserting N elements into an empty linked list, if the linked list needs to be maintained in sorted order? Justify your answer.

(d) Consider the queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q, element) and Dequeue(Q). What is the minimum number of Enqueue operations on Q2 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage? Justify your answer.



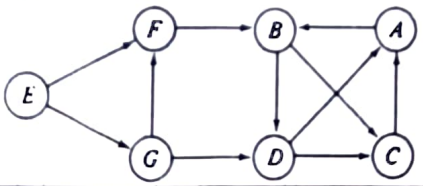
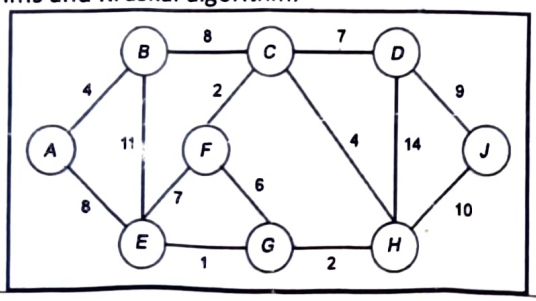


(e) If preorder traversal of a given binary search tree is 15, 10. 12, 11, 20, 18, 16, 19 then find the postorder traversal of the same binary search tree? Justify your answer.

(f) Let G be any connected, weighted, undirected graph:

i. G has a unique minimum spanning tree, if no two edges of G have the same weight.
ii. G has a unique minimum spanning tree, if for every cut G, there is a unique minimum weight edge crossing the cut.

Which of the above two statements is/are TRUE? Justify your answer

	(g)	If a binary search tree ***T*** is constructed using ***N*** unique elements the prove that the time complexity of picking an element in ***T*** that is smaller than the maximum element in ***T*** is $\theta(1)$ i.e., constant time. Use suitable example to prove / justify your answer.	
	(h)	Consider a 13 elements hash table for which f(key)=key mod 13 is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order? Justify your answer	
		UNIT-I	
Q2	How two dimensional arrays are stored in the memory ? Derive the formula for accessing two dimensional array if the base address is B, size of each element is E and the size of array is NxN.	**(10)**	
Q3	Suggest a strategy to check if a string is palindrome or not. Use suitable data structure.	**(10)**	
		UNIT-II	
Q4	Write an algorithm to insert an element into a Doubly Link List at (i) First position (ii) Last Position and (iii) Nth Position.	**(10)**	
Q5	Write an algorithm to print the middle element of Linked List without using any integer variables. Do not assume number of nodes is given.	**(10)**	
		UNIT-III	
Q6	How quick sort and merge sort are different from each other? Compare their complexities and discuss about their suitability for the computer applications.	**(10)**	
Q7	Construct AVL tree for the following data 21,26,30,9,4,14,28,18,15,10,2,3,7 and show the progress of each step clearly in the resultant tree.	**(10)**	
		UNIT-IV	
Q8	How DFS is different from BFS? Show the DFS and BFS for the given below graph, starting at Vertex A. 	**(10)**	
Q9	Consider the given below graph and show the minimum spanning tree using both Prims and Kruskal algorithm. 	**(10)**	

11 12 10 16
19 18 20
15